

# Bram J. Banik

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## Education

**BASc in Engineering Physics** - University of British Columbia

Expected April 2028

- Dean's List - 85.3%
- Selected Courses: Intro to Instrument Design, Differential Equations, Circuit Analysis II, Multivariable/Vector Calculus, Engineering Physics Project I, Signals and Systems

**Westview High School** - Beaverton, OR, USA | GPA: 4.143 | ACT: 36

September 2019 - April 2023

## Skills

**Electrical:** KiCad, PCB Design, Soldering, Filters and Noise Reduction, Debugging, Oscilloscopes

**Mechanical:** Solidworks, Solidworks Simulation, OnShape, ANSYS, CNC Mill, Laser and Waterjet Cutting, Carbon Fiber Wet Layup and Vacuum Infusion

**Software:** Git, C, C++, Java, MATLAB, Python, HTML + CSS, Linux, ESP32 freeRTOS

**ML:** PyTorch, OpenCV, CNNs, NumPy, CUDA C

**General:** Waveforms, Excel, Adobe Illustrator, Adobe After Effects, Notion, Obsidian

## Experience

**UBC Thunderbikes Aerodynamics Team Lead**

August 2025 - Present

- Lead subteam of 10+ students for UBC Thunderbikes first foray into the MotoStudent competition
- Heading projects such as carbon-fiber fairings and custom composite rear frame design
- Managing \$4000 budget for the aerodynamics and composites subteam
- Outlined training for CAD, CNC, and aerodynamics simulations for all-electric racing motorcycle

**Engineering Physics Student Life Coordinator**

July 2025 - Present

- Organizing events for the Engineering Physics student body
- Managing a budget of \$1000+ to distribute among both semesters
- Reigniting interest in social events, with 30+ students attending

**UBC Thunderbikes Aerodynamics Subteam Member**

September 2024 - July 2025

- Designed and built a carbon fiber gas tank and numerous battery panels for an all-electric motorcycle
- Hands-on experience with wet-layup and vacuum infusion processing techniques

## Projects

**Fully Autonomous Pet-Rescuing Robot Competition**

- Engineered an autonomous robot using a computer vision + ESP32 stack
- Trained computer vision model w/ PyTorch, communicating between a Raspberry Pi and an ESP32
- Designed and soldered PCBs for all circuits on the robot, including dual H-Bridges and switch debouncing
- Wrote object-oriented codebase in C++ using freeRTOS to allow efficient time-stitching
- Debugged and handled noise issues from drive motors using shielding and software-based solutions

**SIFT Homography Tracker**

- Developed a GUI used QtDesigner to display live webcam tracking footage
- Used OpenCV's Python library to detect object via the SIFT algorithm and FLANN keypoint matching
- Coded a homography detector when significant keypoints were present using RANSAC

**Linear Regression CNN**

- Implemented a convolutional neural network from scratch in Python
- Integrated backpropagation, gradient descent, and activation functions manually
- Applied CNN to linear regression problem, achieved a 0.7% error from the ideal regression
- Analyzed parameter variation over epochs using matplotlib